

other imposes a tax of t on both markets. The first proposal imposes a high tax rate ($2t$) on a small base (single-family housing), and the second proposal imposes a low tax rate (t) on a large tax base (single-family and condominium housing). The DWL of the first proposal is $(a + b + c + d)$ while the DWL from the second proposal is $(a + a)$, which is half the DWL of the first proposal. Thus, *because DWL goes up “exponentially” as tax rates rise, imposing low tax rates on broader bases typically results in less DWL .* If topics like this are of interest, you should consider taking a course in public finance.

The two proposals also result in different levels of tax revenue. Which proposal actually results in higher revenue for the government? Does this strengthen or weaken the policy proposal “to broaden the base and lower the rates”?

**Exercise
10A.12**

10B

The Mathematics of Consumer Welfare and “Duality”

The mathematical generalization of consumer welfare, as introduced intuitively in part A of this chapter, serves two purposes. First, it allows us, as in concepts introduced in previous chapters, to see how the mathematics of the consumer model can help us generalize the graphical analysis and the intuitions that emerge from it. Second, it provides us a forum in which to bring together all the mathematical techniques introduced so far to paint a full picture of consumer theory, a picture that is commonly referred to as the “duality” of utility maximization and expenditure minimization approaches we have used in the past few chapters.

We will depart in this section from our previous practice of following exactly the same order for our mathematical development in part B of chapters as for the graphical development in part A. Rather, we will begin by demonstrating how the intuitive concepts developed in part A help us generate a full picture of how all the intuitive and mathematical aspects of the consumer model fit together. We do this in Section 10B.1 and then proceed to an application of some of the “duality” mechanism to the topic of consumer welfare, taxation, and deadweight loss.

10B.1 Duality of Utility Maximization and Expenditure Minimization

In previous chapters, we have essentially formulated two different ways of solving optimization problems that (typically) lead to a solution that can be graphed as a tangency between an indifference curve and a budget line. Which of these optimization problems we solved at a particular time depended on what we were trying to answer. Whenever we tried to calculate how much a consumer will actually consume as his or her economic conditions (i.e., the consumer’s income and the prices in the market) change, we solved the utility maximization problem

$$\max_{x_1, x_2} u(x_1, x_2) \text{ subject to } p_1 x_1 + p_2 x_2 = I. \quad (10.1)$$

On the other hand, when we attempted to see how much a consumer will change his or her consumption as prices change while being compensated to keep his or her utility from changing, we solved the expenditure minimization problem

$$\min_{x_1, x_2} E = p_1 x_1 + p_2 x_2 \text{ subject to } u(x_1, x_2) = u \quad (10.2)$$

that gave us the least expenditure necessary for the consumer to reach the same indifference curve u as prices change.

We already showed in Chapter 9 how the solution to problem (10.1) can be written as the (uncompensated) *demand functions*

$$x_1 = x_1(p_1, p_2, I) \text{ and } x_2 = x_2(p_1, p_2, I) \quad (10.3)$$

and how (inverse) slices of these demand functions are related to the various demand curves we have derived graphically. Now consider the solution to problem (10.2), which can be written as

$$x_1 = h_1(p_1, p_2, u) \text{ and } x_2 = h_2(p_1, p_2, u). \quad (10.4)$$

These functions tell us, for any set of prices, how much a consumer will consume of each good *assuming that the consumer is given just enough money to be able to reach utility level u* . For this reason, the functions given in expression (10.4) are often referred to as *compensated demand functions*. They are also known as *Hicksian demand functions* after the economist John Hicks whose work originally identified them, and it is in his honor that we denote the functions in expression (10.4) with “ h ” to distinguish them from the (uncompensated) demand functions in expression (10.3).

10B.1.1 Compensated (or Hicksian) Demand and MWTP In Graph 10.2, we demonstrated how we can derive *MWTP* curves by tracing out the quantity of a good that a consumer would consume at different prices *assuming the consumer gets sufficient compensation to always reach the same indifference curve*. This is exactly what problem (10.2) formalizes mathematically, and the compensated demand functions in expression (10.4) are therefore a simple generalization of the *MWTP* curve derived in Graph 10.2. In fact, when we discussed this derivation in Section 10A.1.2, we mentioned that *MWTP* curves are sometimes referred to as compensated demand curves.

More precisely, note that compensated demand functions are functions of prices and utility. Consider the function $h_1(p_1, p_2', u^A)$ with p_2' set to 1 (as we would do if good x_2 represents “other consumption” denominated in dollars) and utility fixed at the quantity associated with indifference curve u^A . With the other arguments of the function held fixed, this leaves a function of only p_1 , a function that tells us how the consumer will change his or her consumption of x_1 as p_1 changes assuming the consumer is compensated sufficiently to permit him or her to reach indifference curve u^A . The inverse of this function is what is derived graphically in the lower panel of Graph 10.2, the marginal willingness to pay curve associated with the indifference curve u^A . The fact that there exist many *MWTP* curves as demonstrated in Graph 10.4b, one corresponding to each indifference curve, then falls straight out of the underlying mathematics: As different utility levels are plugged into the compensated demand function (instead of u^A), different *MWTP* (or compensated demand) curves emerge.

Exercise 10B.1

In Graph 10.5b, we illustrated that *MWTP* curves and own-price demand curves are the same when tastes are quasilinear. Suppose tastes can be modeled with the quasilinear utility function $u(x_1, x_2) = \alpha \ln x_1 + x_2$. Verify a generalization of the intuition from Graph 10.5b that demand functions and compensated demand functions are identical for x_1 in this case.

10B.1.2 Linking Indirect Utility and Expenditure Functions in the Duality Picture

Once we have solved for demand functions (using utility maximization) and compensated demand functions (using expenditure minimization), we can formally define two further functions that we have already used in previous chapters without naming them: the *indirect utility function*, which tells us for any set of economic circumstances (i.e., prices and income) how much utility the consumer will achieve if she does the best she can; and the *expenditure function*, which tells us for any prices and utility level how big a money budget is required for the consumer to reach that utility level.

To find the utility level a consumer can attain under different economic circumstances, all we have to do is plug the demand functions (which tell us how much the consumer will consume of each of the goods under different circumstances) into the utility function. The indirect utility function $V(p_1, p_2, I)$ can then simply be written as

$$V(p_1, p_2, I) = u(x_1(p_1, p_2, I), x_2(p_1, p_2, I)). \quad (10.5)$$

Similarly, the money required to reach a particular utility level u under different prices is found simply by multiplying the compensated demands for the goods (which tell us how much of each good a person will consume if the person always gets just enough money to reach the utility level u) by the prices and adding them up; i.e., the expenditure function $E(p_1, p_2, u)$ can be written as

$$E(p_1, p_2, u) = p_1 h_1(p_1, p_2, u) + p_2 h_2(p_1, p_2, u). \quad (10.6)$$

Consider, for example, the case of a Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^{(1-\alpha)}$. The utility maximization and expenditure minimization problems yield demand functions

$$x_1(p_1, p_2, I) = \frac{\alpha I}{p_1} \quad \text{and} \quad x_2(p_1, p_2, I) = \frac{(1 - \alpha)I}{p_2} \quad (10.7)$$

and compensated demand functions

$$\begin{aligned} h_1(p_1, p_2, u) &= \left(\frac{\alpha p_2}{(1 - \alpha) p_1} \right)^{(1-\alpha)} u \\ h_2(p_1, p_2, u) &= \left(\frac{(1 - \alpha) p_1}{\alpha p_2} \right)^\alpha u. \end{aligned} \quad (10.8)$$

Verify the solutions given in equations (10.8).

**Exercise
10B.2**

Plugging (10.7) into the Cobb–Douglas utility function, we get the indirect utility function

$$V(p_1, p_2, I) = \frac{I^\alpha (1 - \alpha)^{(1-\alpha)}}{p_1^\alpha p_2^{(1-\alpha)}}. \quad (10.9)$$

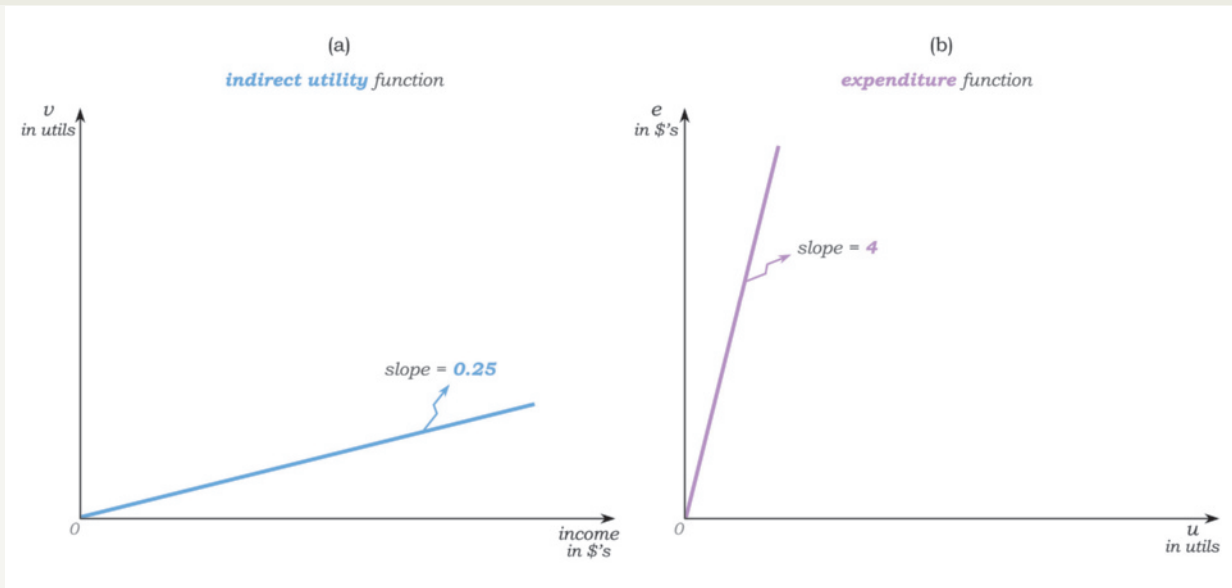
and multiplying the equations in (10.8) by the relevant prices and adding, we get the expenditure function

$$E(p_1, p_2, u) = \frac{u p_1^\alpha p_2^{(1-\alpha)}}{\alpha^\alpha (1 - \alpha)^{(1-\alpha)}}. \quad (10.10)$$

Verify the solutions given in equations (10.9) and (10.10).

**Exercise
10B.3**

Now notice the following: If you set the left-hand side of (10.9) equal to u and solve for I , you get the right-hand side of (10.10). Similarly, if you set the left-hand side of (10.10) equal to I and solve for u , you get the right-hand side of (10.9). That is because *the indirect utility function is the inverse of the expenditure function and vice versa*. Graph 10.11 shows the intuition behind this by graphing first the indirect utility as a function of income when $p_1 = 4$, $p_2 = 1$, and $\alpha = 0.5$, and then graphing the expenditure function (evaluated at the same prices and the same α) as a function

Graph 10.11: Indirect Utility and Expenditure Function

of utility. The only difference between the two graphs is that we have inverted the first graph to get the second, switching the utility and dollar axes in the process!

Other linkages between the utility maximization and the expenditure minimization results can also be identified and should make intuitive sense once you have fully internalized what these functions represent. For instance, suppose we plug the expenditure function in for the income variable I in demand functions. Then, rather than letting income be fixed, we have constructed a new demand function that always provides the consumer sufficient income to reach utility level u . But that is precisely the definition of a compensated demand function. As a result, we can establish the following logical relationship:

$$x_i(p_1, p_2, E(p_1, p_2, u)) = h_i(p_1, p_2, u). \quad (10.11)$$

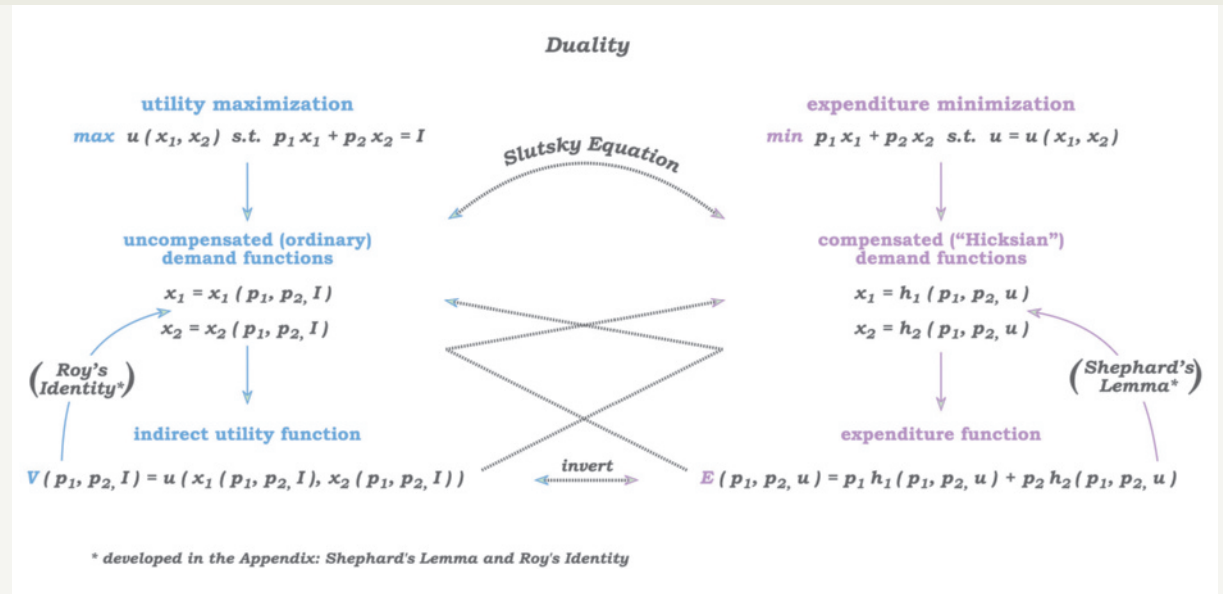
Similarly, suppose we plug the indirect utility function in for the utility term u in compensated demands. Then rather than letting utility be fixed, the compensated demand function would give us the optimal consumption level assuming you have enough income to reach the level of utility you would reach with just income I . In other words, the compensated demand function would then tell you the optimal bundle assuming your income (rather than utility) is fixed, which is just the definition of a regular (or uncompensated) demand function:

$$h_i(p_1, p_2, V(p_1, p_2, I)) = x_i(p_1, p_2, I). \quad (10.12)$$

Exercise 10B.4

Verify that (10.11) and (10.12) are true for the functions that emerge from utility maximization and expenditure minimization when tastes can be modeled by the Cobb–Douglas function $u(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$.

Graph 10.12 summarizes the “duality” picture as we have developed it in this section, and indicates through arrows the linkages between the utility maximization and expenditure minimization problems that we have developed thus far. The arrows labeled “Roy’s Identity” and

Graph 10.12: “Duality” of Utility Maximization and Expenditure Minimization

“Shephard’s Lemma” are developed in the appendix, and the dotted line labeled “Slutsky Equation” is developed next.

10B.1.3 The Slutsky Equation There is one final link between the two sides of our duality picture. It is known as the Slutsky equation, and it relates the slopes (with respect to prices) of uncompensated demand curves to the slopes of compensated demand curves.⁷ To be more precise, we would like to begin with a point that lies on both the demand and the compensated demand functions and then derive the relationship between the slopes of the two functions at that point. We have already done this intuitively in Graphs 10.5 and 10.9. In those graphs, this common point is point A, and it is at that point that we could say which of the two curves is steeper depending on whether the good is normal or inferior. It is easiest for us to begin our mathematical derivation of the Slutsky equation with expression (10.11), which already relates demand functions to compensated demand functions (but does not relate their slopes to one another). To identify the relationship of the slopes, we simply take the partial derivative (with respect to one of the prices) of each side of equation (10.11). This requires us to invoke the chain rule from calculus since the function on the left-hand side contains the expenditure function E that itself is a function of prices:

$$\frac{\partial x_i}{\partial p_j} + \left(\frac{\partial x_i}{\partial E} \right) \left(\frac{\partial E}{\partial p_j} \right) = \frac{\partial h_i}{\partial p_j}. \quad (10.13)$$

Rearranging terms, and replacing the E term in $(\partial x_i / \partial E)$ with I (since expenditure is the same as income in the consumer model), we can write this equation as

$$\frac{\partial x_i}{\partial p_j} = \frac{\partial h_i}{\partial p_j} - \left(\frac{\partial x_i}{\partial I} \right) \left(\frac{\partial E}{\partial p_j} \right). \quad (10.14)$$

⁷The Slutsky Equation is named after the Russian economist and statistician Eugene Slutsky (1880–1948). The equation was so named by John Hicks who also called it the “Fundamental Equation of Value Theory.”

Equation (10.14) is written in terms of good x_i and price p_j . To help us investigate precisely how this equation relates to the intuitions we have developed so far, suppose that we focus on good x_1 and a change in p_1 . Equation (10.14) can then be written as

$$\frac{\partial x_1}{\partial p_1} = \frac{\partial h_1}{\partial p_1} - \left(\frac{\partial x_1}{\partial I} \right) \left(\frac{\partial E}{\partial p_1} \right). \quad (10.15)$$

The left-hand side of equation (10.15) is the change in the actual quantity demanded of good x_1 when p_1 changes marginally. This is analogous to the move from A to C in Graphs 10.5 and 10.8, although the calculus here corresponds to marginal (or very small) changes. The first term on the right-hand side of equation (10.15) is the change in the quantity of x_1 demanded assuming the consumer has been compensated to keep his or her utility constant. It is analogous to the move from A to B in Graphs 10.5 and 10.9, or the substitution effect. This must mean that the final term in equation (10.15) is analogous to the move from B to C in Graphs 10.5 and 10.9, or the income effect. Indeed, that is precisely what the final term suggests: $(\partial x_1 / \partial I)$ is the change in the quantity of x_1 demanded when income changes, and $(\partial E / \partial p_1)$ is the size of the required compensation given that p_1 changes.

First, note that $\partial h_1 / \partial p_1 < 0$; when price increases, the substitution effect always suggests we will purchase less of that good when we are compensated. Now suppose that we know that a consumer's tastes are quasilinear in x_1 ; i.e., x_1 is borderline between a normal and an inferior good. This implies that consumption of x_1 does not change as income changes, or $\partial x_1 / \partial I = 0$, reducing equation (10.15) to

$$\frac{\partial x_1}{\partial p_1} = \frac{\partial h_1}{\partial p_1}. \quad (10.16)$$

This is precisely what is illustrated intuitively in Graph 10.5b, where we demonstrated that demand curves and *MWTP* (or compensated demand) curves are the same for quasilinear goods. The reason for this is that the income effect disappears in this special case, leaving us with only the substitution effect.

Now suppose we knew instead that x_1 was a normal good. In that case, $\partial x_1 / \partial I > 0$. Whenever the price of a good we are consuming goes up, it must furthermore be true that the expenditure required to reach the same utility level increases, thus $\partial E / \partial p_1 > 0$. Together, these two statements imply that the second term in equation (10.15) is negative (two positive terms multiplied by each other and preceded by a negative sign). Thus, when x_1 is a normal good, the quantity demanded falls first because of the substitution effect ($\partial h_1 / \partial p_1 < 0$) and then again because of the income effect ($-(\partial x_1 / \partial I)(\partial E / \partial p_1) < 0$). When x_1 is an inferior good, on the other hand, $\partial x_1 / \partial I < 0$, which implies that the second term on the right-hand side of equation (10.15) is positive. Thus, income and substitution effects point in opposite directions. All this is precisely as we have concluded in our graphs of consumer choices.

10B.1.4 Graphs and Inverse Graphs Sometimes students get confused when looking at a graph like the lower panel of Graph 10.5 and attempting to relate the slopes of the demand and *MWTP* curves in the graph to the slopes represented by partial derivatives in equation (10.15). For instance, suppose again that x_1 is a normal good. Then it appears that the slope of the demand curve ($\partial x_1 / \partial p_1$) is negative because of the negative slope of the *MWTP* curve ($\partial h_1 / \partial p_1$) and because of an additional negative component implicit in the second term of equation (10.15), the income effect. This would mean that the slope of the demand curve at any point is a negative number that is larger in absolute value than the slope of the *MWTP* curve at that same point. Put differently, it means that the demand curve is downward sloping and steeper than the *MWTP* curve (which is also downward sloping). But Graph 10.5a suggests the opposite, that the demand curve is downward sloping and *shallower* than the *MWTP* curve for a normal good.

The reason for the appearance of a discrepancy between the intuition developed in Graph 10.5a and the math implicit in equation (10.15) can once again be found in the unfortunate fact that economists graph demand curves as slices of *inverse* demand functions. Thus, the slopes derived from the mathematics represent the inverse of the slopes derived in our graphs. Taking an inverse of a slope does not change the sign of that slope (i.e., downward-sloping curves remain downward sloping), but it does change whether one curve is relatively steeper than the other (i.e., steep slopes become shallow slopes and vice versa). Graph 10.13 illustrates this relationship by plotting demand curves as inverse slices of demand functions in panel (a) (which illustrate demand and *MWTP* curves as depicted in Graph 10.9) and as simple slices of the same demand functions (with the axes reversed) in panel (b). The arrows in each graph begin with the demand curve representing a normal good and end with the demand curve representing a Giffen good. The slopes in the Slutsky equation correspond to the slopes in the second graph.

10B.2 Taxes, Deadweight Losses, and Consumer Welfare

As suggested in Section 10A, concepts like consumer surplus and deadweight loss can be read off as distances in the consumer diagram or as areas below *MWTP* curves. Areas under curves can be calculated mathematically as integrals, but we do not have to resort to integral calculus to be able to calculate changes in consumer surplus or deadweight loss mathematically. This section thus uses the same example of a housing tax discussed throughout Section A to demonstrate how the relevant concepts can be calculated without any additional calculus and simply using the various parts of our duality picture. (The more mathematically inclined students can turn to the end of the appendix to see an explanation of how areas along compensated demand curves correspond to distances in the consumer diagram.)

10B.2.1 Using Duality Concepts to Calculate Deadweight Loss We concluded in Section 10A.3 that taxes are inefficient because of substitution effects. We furthermore defined the size of the inefficiency through a measure of deadweight loss (*DWL*)—the difference

Graph 10.13: Inverse Demand and Demand for Different Types of Goods

